

standards and protocols (ISO, EPC)
13. Regional (US, European and Asian) regulations

14. Will the tag need to store more than just an ID number like an EPC?
15. Anti-collision—how many tags in

TABLE IV
Managing the Problem of Tag Orientation

Reader-antenna	Tag
• UHF—Antenna circular field polarisation • HF—Antennae physically placed at different locations with different orientations (XYZ) • Two antennae 90° out of phase with each other • 3D tunnel readers	• UHF—Two antennae polarised 90° out of phase, e.g., Matrics double dipole

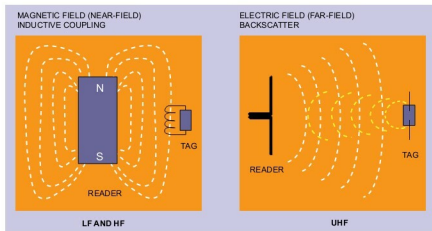


Fig. 6: Two different ways of energy and information transfer between the reader and tag

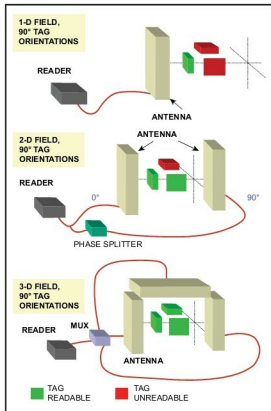


Fig. 7: HF tag orientation with different antenna configurations

the field must be detected at the same time and how quickly?

16. How fast will the tags move through the reader field?

17. Reader support—which reader products are able to read the tag?

18. Does the tag need to have security?

How tags communicate

In order to receive energy and communicate with a reader, passive tags use one of the two following methods shown in Fig. 6. These are near-field, which employs inductive coupling of the tag to the magnetic field circulating around the reader antenna (like a transformer), and far-field, which uses techniques similar to radar (backscatter reflection) by coupling with the electric field.

The near-field is generally used by RFID systems operating in the LF and HF bands, and the far field is used for longer-range UHF and microwave RFID systems. The theoretical boundary between the two fields depends on the frequency used, and is in fact directly proportional to $\lambda/2\pi$, where ' λ ' is wavelength. This gives, for example, around 3.5 metres for a HF system and 5 cm for UHF, both of which are further reduced when other factors are taken into account.

Tag orientation (polarisation)

How tags are placed with respect to the polarisation of the reader's field can have a significant impact on the communication distance for both HF and UHF tags. This can result in a 50 per cent reduction of the operating range and, in the case of the tag being displaced by 90° (see Fig. 7), inability to read the tag.

The optimal orientation of HF tags is when the two antenna coils (reader and tag) are parallel to each other as shown in Fig. 7. UHF tags are even more sensitive to polarisation due to the directional nature of the dipole fields. The problem of polarisation can be overcome to a large extent by different techniques implemented either at the reader or tag as shown in Table IV.

The future

Developments in RFID technology continue to yield larger memory capacities, wider reading ranges and faster processing. However, it is highly unlikely that the technology will ultimately replace barcode. Even with the inevitable reduction in raw materials coupled with economies of scale, the integrated circuit in an RF tag will never be as cost-effective as a barcode label. RFID, though, will continue to grow in its established niches where barcode or other optical technologies are ineffective, such as in the chemical container and livestock industries. ●

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